Maths 4

18 March 2026 00:24

- Find GCD
- Find LCM using GCD
- Pascal's Triangle → later
- Output:
```
char ch = 'A';
int x = ch;
System.out.println(x);
```
- Output:
```
System.out.println('A' + 1);
```
- Divide 10 by 3 and roundoff the answer upto 3 decimal Places.

} Type Conversion

HCF (25, 35)

→ Highest common factor.

5 → HCF

25 → 1, 5, 25

35 → 1, 5, 35

HCF (25, 35)

1 — 35

Highest

1, 2, 3, 4, 5, 6, 7 35

Common factor

Common factor

Divide

$\frac{25}{1}$, $\frac{35}{1}$

ans = 1

ans = 1

ans = ~~1~~ 5

i = 25

$\frac{25}{25}$, $\frac{35}{25}$

i = 26

$\frac{25}{5}$, $\frac{35}{5}$

26

3

25, 35

```
int n = 25, m = 35 ; ans = 0;
for (int i = 1 ; Math.min(m, n); i++){
    if (n % i == 0 && m % i == 0) ans = i;
```

1 → ~~35~~ 25

}

sout(ans);

25) 35 (1
25
10) 25 (2
20
5) 10 (2
10
0

HCF(25, 35)

= HCF(10, 25)

= HCF(5, 10)

Euclidian Method:

25, 35

25) 35 (
25
10) 25 (2
20
5) 10 (2
10
0

HCF

HCF (12, 18)

12) 18 (1
12
6) 12 (2
12
0) 6 (

HCF:

(0, 6)

HCF

29/10

0, 1, 2 ----- 9

< 10

HCF (18, 190)

~~18) 190 (1~~
~~18~~

10) 18 (1
10
8) 10 (1
8
2) 8 (4
8
0

HCF

HCF (18, 190)

= HCF (10, 18)

= HCF (8, 10);

= HCF (2, 8) //

HCF (2, 0)

```
int HCF () {
    int n=25, m=35;
    while (n>0 && m>0) {
        if(n>m) n = n%m;
        else m = m%n;
    }
    if (m==0) Sout (n);
    else sout(m);
}
```

$m \times n = \boxed{LCM(m,n)} * HCF(m,n);$

$\frac{m \times n}{HCF(m,n)} = LCM(m,n);$

```
int n=25, m=35
int Lcm = 0;
Lcm = (m×n) / HCF(m,n);
Sout (Lcm);
```

Q 3.

Print.

1 2 1

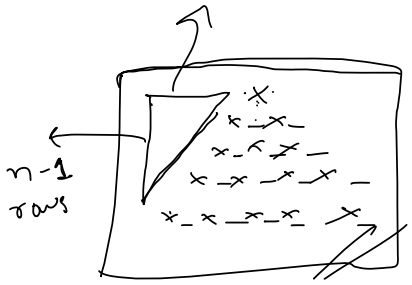
1 1 2 2 1

1 4 6 4 1

→ 1 2 1

→ 1 3 3 1

→ 1 4 6 4 1



row → x x x x x x

n-5

n-i

1 → n-i+1

0 → n-i

```
for (int i=0 ; i<n ; i++){
    for (int j=0 ; j< n-i ; j++){
        Sout(" ");
    }
    for (int j=0 ; j<i ; j++){
        Sout("* "+" ");
    }
```

n=3

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

1 2 1

nth row

$$\boxed{{}^{n}C_{r}} = \frac{n!}{(n-r)!\,r!}$$

row-1 C col-1

$$\frac{n!}{(n-r)!\,r!}$$

$(n-3) \times \cdots 1$

$n \times (n-1) \times (n-2) \ldots$

$(n-r) \times (n-r-1) \times (n-r-2) \ldots \times 2$

factorial (int n)

$5! / 3! \times 2!$

$$^5C_2 = \frac{5!}{(5-2)!\,2!} = \frac{5!}{3! \times 2!} = \frac{5 \times 4 \times (3 \times 2 \times 1)}{3! \times 2!}$$

$$\frac{5 \times 4 \times 3!}{3! \times 2!} = \frac{5 \times 4}{2 \times 1} = \frac{5}{1} \times \frac{4}{2}$$

$$\frac{5-0}{1} \times \frac{5-1}{2}$$

$$\frac{5-0}{0+1} \times \frac{5-1}{1+1}$$

$n = 5$

for (int i = 0; i < n; i++){

$$ans = ans \times \frac{n-i}{i+1};$$

$$\frac{5!}{3! \times 2!}$$

$$\frac{5 \times 4 \times 3!}{3! \times 2!}$$

1×2

$$\frac{5-1}{1+1}$$

$4/2$

$$\frac{5}{1} \times \frac{4}{2}$$

$$\frac{5 \times 4}{1 \times 2}$$

for $i = 0$

$$\frac{5-0}{0+1} = \frac{5}{1}$$

$$q = \frac{5-1}{1+1} = \frac{4}{2}$$

$$5! = \frac{5 \times 4 \times (3 \times 2 \times 1)}{2 \times 1 \times 2!} \rightarrow 35$$

1. $^{10}C_3$ 2. $^{7}C_4$ 3. $^{9}C_2$ →

4. $^{12}C_9$ 5. $^{18}C_6$ → 18564

↳ $^{n}C_r = \frac{n!}{(n-r)!\,r!}$

int — data type

float — ↳ decimal places

long double

```
long a = 5;
int b = 6;
b = a;
```

→ b = a

a=b (✓).

long a = 2

long a = 6 (< int_range

int b = 5

b = (int) a ; → explicit type conversion

1 + 1.0

→ int

→ float datatype

2.0 → float

int a = 2

int b =

Sout ((fl

(float) (2/3)

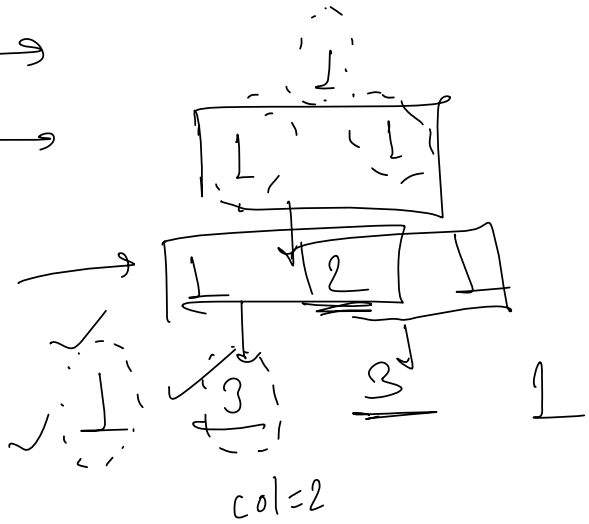
(float) 0 →

1st row →

2nd →

3rd →

yth →



yth row

row-1 C col-

row-1 C col-1

$^{n}C_{r} = \frac{n!}{(n-r)! \times r!}$

$^{3}C_{0}$

$^{3}C_{2}$

$^{10}C_{3} = {}^{n}C_{r} = \frac{n!}{(n-r)! \times r!} =$

$n! = 1 \times 2 \times 3 \cdots \times n$

$$= \frac{10 \times 9 \times \overset{8}{\cancel{7}} \times \cancel{6 \cdots 1}}{\cancel{7 \times 6 \times 5 \times \cdots 1} \times 3!}$$

$$= \frac{10 \times \overset{3}{\cancel{9}} \times \overset{4}{\cancel{8}}}{\cancel{3} \times \cancel{2} \times 1}$$

$$= 120 \text{ Ans.}$$

1

1 1

1 2 1

4th row = 1 3 3 1

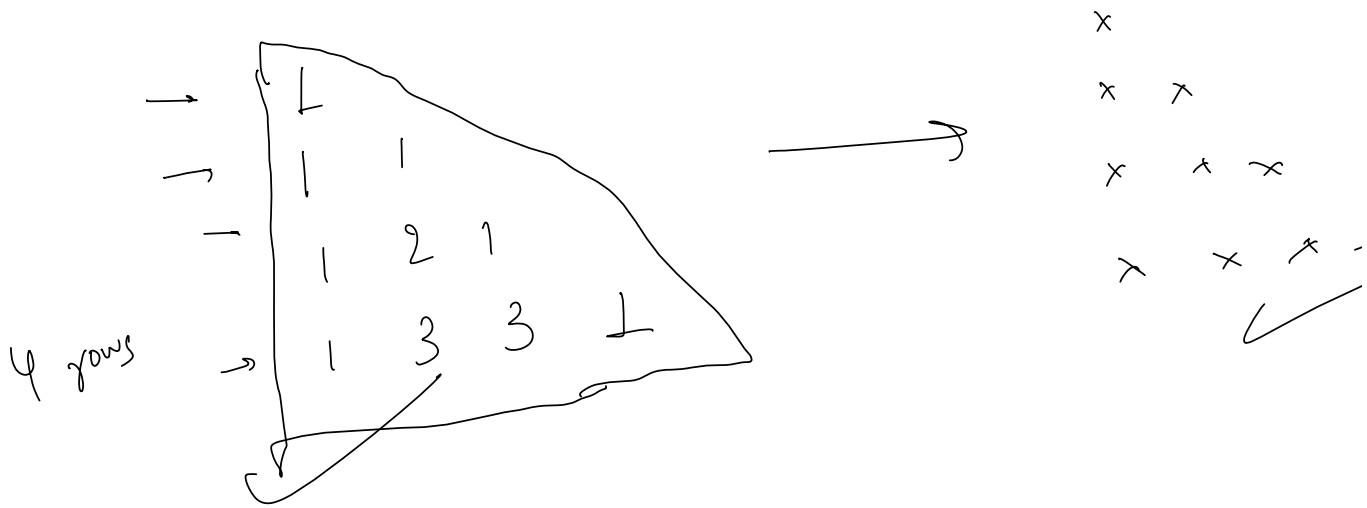
Imp!! $^{row-1}C_{col-1}$

$$^{4-1}C_{1-1} = {}^{3}C_{0} = 1$$

$$^{4-1}C_{2-1} = {}^{3}C_{1} = 3$$

$$^{4-1}C_{3-1} = {}^{3}C_{2} = 3$$

$$^{4-1}C_{4-1} = {}^{3}C_{3} = 1$$




```
for(int i=0; i<n; i++){
    for(int j=0; j<i; j++){
```

$^{5}C_{3} =$

```
for(int i=1 → n){
    ans* = i;
}
```

$=$

$=$

Sol

$n = 5$

$$\boxed{\frac{5-0}{0+1} \times \frac{5-1}{1+1}} \times \cancel{\frac{5-2}{2+1}}$$

$$^{10}C_3 = \frac{10 \times 9 \times 8 \times 7}{7! \times 3}$$

$n = 10$

$$= \frac{10 \times 9 \times 8}{1 \times 2 \times 3}$$

$$= \boxed{\frac{10-0}{0+1} \times \frac{10-1}{1+1} \times \frac{10-2}{2+1}}$$

```
for (int i = 0 ; i<n ; i++){
    int ans = 1;
    for(int j = 0; j<=i ; j++){
        Sout (ans);
        ans x = (i-j)/(j+1);
    }
```

cout ("\n");

i = 0

i = 1

i = 2

i = 3

1 2 1

i = 4

1 3 3 1

n = 4

row-1 C col-1

n = 4 = 1 3 3 1

~~for (int i = 0 ; i < n ; i++) {~~

nth

n = 4

→ 1 3 3 1

int ans = 1 ;

for (int i = 0 ; i < 4 ; i++) {

sout (ans);

$$ans = ans \times \boxed{\left(\frac{n-i}{i+1}\right)}$$

$$^{10}C_3 = \frac{10 \times 9 \times 8 \times \cancel{7!}}{\cancel{7!} \times 3!}$$

$$= \frac{10 \times 9 \times 8}{1 \times 2 \times 3} =$$

$i = 0$, $ans = 1$

1 3 3 1

$i = 1$, $ans = ans \times \frac{4-1}{0+1}$

$i = 2$ $ans = 3 \times \frac{4-2}{0+2}$

$i = 3$ $ans = 3 \times \frac{4-3}{0+3}$